MITRC

MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE

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Subject & Subject Code: 6CS5-13: Natural Language Processing

Semester: VI

Session: 2023-24(Even Sem.)

Unit No.: 3

Date of Submission:13.03.2024

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14/3/2024

DEAN ACADEMICS

UNIT-3 Semantic Analysis

NLP - Meaning Representation

LECTURE NO:-1

Semantic analysis is a **branch of general linguistics** which is the process of understanding the meaning of the text. The process enables computers to identify and make sense of documents, paragraphs, sentences, and words as a whole.

Semantic analysis is done by **analyzing** the grammatical structure of a piece of text and understanding how one word in a sentence is related to another.

INTRODUCTION

- Semantic analysis deals with **analyzing the meanings** of words, fixed expressions, whole sentences, and utterances in context. In practice, this means **translating original expressions** into some kind of semantic metalanguage.
- **NLP applications** of semantic analysis for long-form extended texts include information retrieval, information extraction, text summarization, data-mining, and machine translation and translation aids.
- Semantic analysis is also pertinent for much shorter texts and handles right down to the single-word level. These cases arise in examples like understanding user queries and matching user requirements to available data.
- The elements of semantic analysis are also of high **relevance** in efforts to improve web ontologies and knowledge representation systems.

WHAT IS SEMANTIC ANALYSIS

Semantic analysis is the process of **drawing meaning from text** and it allows computers to understand and interpret sentences, paragraphs, or whole documents by **analyzing their grammatical structure**, and identifying relationships between individual words in a particular context.

- Semantic analysis **captures the real meaning** of a given paragraph by first reading it and then identifying the text elements and their grammatical role.
    - It can identify **whether a text is based on a certain category** like sports, or a category, like politics, based on the words in the text.
    - Even if the related words are not present, the analysis can still identify what the text is about.

Lexical semantics plays an important role in semantic analysis, allowing machines to understand relationships between lexical items like words, phrasal verbs, etc.

Parts of Semantic Analysis

The semantic analysis focuses on larger chunks of text, whereas lexical analysis is based on smaller tokens.

- As we focus on larger chunks in semantic analysis, we **divide the semantic analysis** into two parts:
    - **Studying the meaning of the Individual Word**: This is the first component of semantic analysis in which we study the meaning of individual words. This component is known as lexical semantics.
    - **Studying the combination of Individual Words**: In this component, we combined the individual words to provide meaning in sentences.

Tasks Involved in Semantic Analysis

- The most important task of semantic analysis is to find the proper meaning of the sentence using the elements of semantic analysis in NLP.
- There are various other **sub-tasks** involved in a semantic-based approach for machine learning, including word sense disambiguation and relationship extraction.

2. Relationship Extraction

Relationship extraction is the task of detecting the semantic relationships present in a text. Relationships usually involve two or more entities which can be names of people, places, company names, etc. These entities are connected through a semantic category such as works at, lives in, is the CEO of, headquartered at etc.

- Rule-based RE, Weakly Supervised RE, Supervised RE, Distantly Supervised RE, and Unsupervised RE are some methods generally used for relationship extraction (RE)
- Relation Extraction is a key component for building relation knowledge graphs and also of crucial significance to natural language processing applications such as structured search, sentiment analysis, question answering, and summarization.
- Relation Classification, Joint Entity, Relation Extraction, Temporal and Dialog Relation Extraction, Binary and Continual Relation Extraction, Hyper-Relational Extraction, Multi-Labeled Relation Extraction, and Relation Mention Extraction are some **sub-tasks involved in relation extraction** under elements of semantic analysis.
- Relationship extraction algorithms can be designed using either **open form or targeted form**.

- **Open relationship extraction**: The algorithm finds and returns the relationship's text snippets along with its arguments. Most of the relationship extraction tools in today's marketplace utilize open relationship algorithms to perform the task.
    - These methods produce semi-structured results and need some human interpretation.
    - There is also **no constraint** as it is not limited to a specific set of relationship types.
- **Targeted relationship extraction**: In this type, we pre-train the algorithms for identifying specific relationship types.
    - Targeted relationship extraction algorithms produce **structured information**, and downstream applications like the knowledge graph easily digest it.
    - These algorithms typically **extract relations by using machine learning models** for identifying particular actions that connect entities and other related information in a sentence.

Relationship extraction involves first identifying various entities present in the sentence and then extracting the relationships between those entities.

ELEMENTS OF SEMANTIC ANALYSIS

Studying a language **cannot be separated from studying the meaning of that language** because when one is learning a language, we are also learning the meaning of the language.

Let us learn about the difficulties associated with learning the meaning and classification of different kinds of relations that need to learn under elements of semantic analysis.

Difficulties with Understanding the Meaning of Language

- The meaning of a language can be seen from its relation between words, in the sense of how one word is related to the sense of another. These relations can be studied under the domain of sense relations.
- Sense relations can be seen as revelatory of the semantic structure of the lexicon.
- Sense relations are the relations of meaning between words as expressed in hyponymy, homonymy, synonymy, antonymy, polysemy, and meronymy which we will learn about further.

Classification of Sense Relations

- Sense relations can be classified into two classes - Those that express identity and inclusion between word meanings and those that express opposition and exclusion.
    - The first class discusses the sense relations between words whose meanings are similar or included in other ones.

- The second class discusses the sense relations between words whose meanings are opposite or excluded from other words.

- Sense relations can also be classified into two groups of horizontal and vertical sense relations under elements of semantic analysis.
  - Horizontal relations include synonymy, which is the sameness of meaning of different linguistic forms, and also various other relations that can be subsumed under the general notion of opposition.
    - Synonymy is the sameness of the meaning of different linguistic forms, for example, orange and apfelsine in German with the same meaning of orange.
  - The two principal vertical relations are **hyponymy and meronymy**.
    - Other than these two principal vertical relations, there is another vertical sense relation for the verbal lexicon used in some dictionaries called **troponymy**.
    - **Troponymy** refers to the case where one verb specifies a certain manner of carrying out the activities referred to by the other verb, such as in amble -- walk or shelve -- put.
- **Horizontal sense relations of Opposition**: The oppositions forms include incompatibility, antonymy, complementarity, conversity, and reversity.
  - **Incompatibility** is mutually exclusive in a particular context, for example, fin--foot.
  - **Antonymy** is gradable, directionally opposed words, for example, big--small.
  - **Complementarity** is the exhaustive division of conceptual domain into mutually exclusive compartments like aunt--uncle, possible--impossible
  - **Conversity** are static directional opposites with the specification of one lexical element in relation to another along some axis, for example above--below
  - **Reversity** are dynamic directional opposites with motion or change in opposite ways like ascend--descend.

Hyponymy

Hyponymy is the case when a relationship between two words, in which the **meaning** of one of the words **includes** the meaning of the **other** word.

- In hyponymy, the meaning of one lexical element hyponym is **more specific than the meaning of the other word**, which is called hyperonym under elements of semantic analysis.
- Example: Relation between cat and animal, pigeon and bird, orchid and flower.
  - Cat is said to be a hyponym of animal, pigeon a hyponym of bird, and orchid a hyponym of the flower.

- Animal is said to be the superordinate which is also called the hyperonym of cat, bird the superordinate of pigeon, and flower the superordinate of orchid.
- This relation is also described **as one of inclusion**.

Sample example illustration for hyponymy

- From the flow chart we can see that the class denoted by the **superordinate term** includes the class denoted by the **hyponym as a sub-class**.
- The class of flower includes the class of orchid as one of its **subclasses**.
- Flower is the general term that includes orchids and other types of flowers such as roses and tulips.
- The relationship between the orchid rose, and tulip is also called **co-hyponym**.

Homonymy

Homonymy refers to the case when words are written in the same way and **sound alike** but have **different meanings**.

- Example: Our house is on the west bank of the river. vs. I want to save my first salary in the bank.
  - Bank in both these sentences are homonyms and are written in the same way and sound alike but their meanings are different.
  - In the first example, the bank refers to the side of a river and the land near it while in the second, it is an organization that provides various financial services.
- In **relation to lexical ambiguities**, homonymy is the case where different words are within the same form, either in sound or writing. For example, light (vs. dark) and light (vs. heavy).

Homonymy refers to two or more lexical terms with the same spellings but completely distinct in meaning under elements of semantic analysis.

Synonymy

Synonymy is the case where **a word which has the same sense or nearly the same as another word**.

- Example: happy, content, ecstatic, overjoyed which convey the same emotion of happiness.
- It can also be seen as the case when two or more lexical terms that might be **spelled distinctly have the same or similar meaning**.

A pair of words can be **synonymous in one context** but may be **not synonymous in other contexts** under elements of semantic analysis.

- We come across many **cases where words are synonymous in various places** like The street is very wide/deep.; You have my deep/profound sympathy.
    - In both of these sentences, wide and broad and deep and profound are replaceable to each other, and the meaning of the sentence remains the same so we can replace each other.
- We also have cases where there are **synonymous in one context but not in the other one**.
    - The student speaks with a broad British accent vs. The student speaks with a wide British accent.
    - The pair of wide and broad in this sentence is not synonymous since wide cannot substitute broad in that context.

Antonymy

Antonyms refer to pairs of lexical terms that have contrasting meanings or words that have close to opposite meanings.

- **Example:** long: short, fast: slow, easy: difficult, good: bad, hot: cold, etc.
- There are three types of antonyms: Polar antonyms, Equipollent antonyms, and Overlapping Antonyms.
- **Polar Antonyms**: This type of oppositeness can be easily modified with degree modifiers such as very, rather, quite, slightly, extremely, and the like.
    - Example: The essay is very long.; The test is quite easy.; His performance is not extremely bad.
    - Polar antonyms can also be applied in the **comparative or superlative degree** as in easy – easier – easiest; bad – worse – worst.
    - One other characteristic of this antonym is that a **normal how to question** can be applied only to one member of a pair.
    - Example: How long is your essay?; But not?; How short is your essay?
- **Equipollent Antonyms**: This type of antonym refer to **subjective sensations** as hot: cold, bitter: sweet, painful: pleasurable, or emotions as happy: sad, proud of, ashamed of.
    - The characteristic of these antonyms is that **normal how-question** can be applied to both terms of the pair.

- How hot is the weather? vs. How cold is the weather?

- **Overlapping Antonyms**: All overlapping antonym pairs have an evaluative polarity as part of their meaning as good: bad, kind: cruel, clever: dull, pretty: plain, polite: rude.
  - The how-question can be applied to both terms of the pairs, but one term yields a **neutral question and the other one a committed question**.
  - Example: How good is the performance? vs. How bad is your test score?

Polysemy

Polysemy is defined as **word having two or more closely related meanings**. It is also sometimes **difficult to distinguish homonymy from polysemy** because the latter also deals with a pair of words that are written and pronounced in the same way.

- It **differs** from homonymy because the meanings of the terms **need not be closely related** in the case of homonymy under elements of semantic analysis.
- Example: The sun is very bright today. vs. She is a very bright student.
  - In the first sentence, bright means shining, and in the second, it means intelligent. The related meanings are that the sense of a bright student brings a sense of shine in that a bright student usually shines their intelligence.
- Polysemy **poses greater problems than homonym** because the meaning differences concerned and the associated syntactic and other formal differences are typically **more subtle**.
- **Mishandling** of polysemy is a **common failing** of semantic analysis, both the positing of false polysemy and failure to recognize real polysemy.
  - The problem of **false polysemy** is very common in conventional dictionaries like Longman, WordNet, etc.
  - The problem of **failure to recognize polysemy** is more common in theoretical semantics, where theorists are often reluctant to face up to the complexities of lexical meanings.

Polysemy refers to a relationship between the meanings of words or phrases, although slightly different, and shares a common core meaning under elements of semantic analysis.

Difference between Polysemy and Homonymy

- Both polysemy and homonymy words have the same syntax or spelling. The main difference between them is that in polysemy, the meanings of the words are related, but in homonymy, the meanings of the words are not related.
- **Example:** If we talk about the same word bank, we can write the meaning as a financial institution or a river bank. In that case, it would be an example of a homonym because the meanings are unrelated to each other.

Homonymy and polysemy deal with the **closeness or relatedness** of the senses between words. Homonymy deals with different meanings and polysemy deals with related meanings.

Meronymy

Meronymy is when the meaning of one lexical element specifies that its **referent is part of the referent of another lexical element**. Example: hand -- body.

- Meronomy is also a logical arrangement of text and words that denotes a constituent part of or member of something under elements of semantic analysis.

Meronomy refers to a relationship wherein one lexical term is a constituent of some larger entity like Wheel is a meronym of Automobile.

CONCLUSION

- Semantic analysis helps in understanding the context of any text and understanding the emotions that might be depicted in the sentence.
- Words with multiple meanings in different contexts are ambiguous words, and word sense disambiguation is the process of finding the exact sense of them.
- Hyponym relations refer to the super and subordinate relationships between words, earlier called hypernyms and later hyponyms.
- Synonymy refers to word which has the same sense, and antonymy refers to words that have contrasting meanings under elements of semantic analysis.
- Polysemy is the coexistence of many possible meanings for a word or phrase, and homonymy is the existence of two or more words having the same spelling or pronunciation but different meanings and origins.

Meaning Representation,

The semantic analysis creates a representation of the meaning of a sentence. But before deep dive into the concept and approaches related to meaning representation, firstly we have to understand the building blocks of the semantic system.

Building Blocks of Semantic System

While representing the meaning of the words, the following building blocks play an important role –

Entities

It represents the individual such as a particular organization, location, people's name, etc.

For Example,

Punjab, China, Chirag, Kshitiz all are entities.

Concepts

It represents the general category of the individuals such as a person, city, etc.

Relations

It represents the relationship between entities and concepts.

For Example,

Sentence: Ram is a person

Predicates

It represents the verb structures.

For Example,

Semantic roles and Case Grammar

Now, we have a brief idea of meaning representation that shows how to put together the building blocks of semantic systems. In other words, it shows how to put together entities, concepts, relations, and predicates to describe a situation. It also enables reasoning about the semantic world.

Approaches to Meaning Representations

The semantic analysis uses the following approaches for the representation of meaning −

- First-order predicate logic (FOPL)
- Semantic Nets
- Frames
- Conceptual dependency (CD)
- Rule-based architecture
- Case Grammar
- Conceptual Graphs

Need of Meaning Representations

The reasons behind the need for the meaning representation are as follows:

Linking of linguistic elements to non-linguistic elements

With the help of meaning representation, we can link linguistic elements to non-linguistic elements.

Representing variety at the lexical level

With the help of meaning representation, we can represent unambiguously, canonical forms at the lexical level.

Can be used for reasoning

The meaning representation can be used to reason for verifying what is correct in the world as well as to extract the knowledge with the help of semantic representation.

UNIT-3 Semantic Analysis

NLP - Lexical Semantics

LECTURE NO:-2

It is the first part of semantic analysis, in which we study the meaning of individual words. It involves words, sub-words, affixes (sub-units), compound words, and phrases also. All the words, sub-words, etc. are collectively known as lexical items.

In simple words, we can say that lexical semantics represents the relationship between lexical items, the meaning of sentences, and the syntax of the sentence.

The steps which we have to follow while doing lexical semantics are as follows:

- Classification of lexical items.
- Decomposition of lexical items.
- Differences, as well as similarities between various lexical-semantic structures, are also analyzed.

LEXICAL SEMANTICS

It's the initial step in semantic analysis, where we look at the meaning of individual words. All included words, sub-words, affixes (sub-units), compound words, and phrases. All words, sub-words, and other lexical items are lexical items. In simple terms, lexical semantics is the link between lexical elements, sentence meaning, and sentence structure.

The following are the stages that must be followed when performing lexical semantics:

- Lexical item classification.
- Lexical item decomposition.

There are also comparisons and differences between distinct lexical-semantic structures.

Example:

Let's use the word forward as an example.

Let's have a look at two sentences.

"Forward into the ocean," says the narrator.

"She was leaning forward."

In both of these circumstances, we can split forward.

The words' Forward' and 'forward' have two different meanings from other words. "Forward into the ocean," says the narrator. It is a statement in which the forward refers to the word 'ocean,' which is connected. "She was leaning forward." This, on the other hand, is a past tense action and refers to 'she.'

These are distinct based on the English language, but how are these semantics represented? How can we tell the difference between these two?

Other words can be seen as being related to that one term.

UNIT-3 Semantic Analysis

NLP - Ambiguity,

LECTURE NO:-2

Ambiguity in Natural Language Processing

WHAT IS AMBIGUITY IN NATURAL LANGUAGE PROCESSING? TYPES OF AMBIGUITY IN NLP. WHICH STEP OF NLP IS AMBIGUOUS? DEFINE AMBIGUITY IN NLP. EXAMPLES OF AMBIGUITY WITH EXPLANATION.

AMBIGUITY

Ambiguity in computational linguistics is a situation where a word or a sentence may have more than one meaning. That is, a sentence may be interpreted in more than one way. This leads to uncertainty in choosing the right meaning of a sentence especially while processing natural languages by computer.

- Ambiguity is a challenging task in natural language understanding (NLU).
- The process of handling the ambiguity is called as disambiguation.
- Ambiguity presents in almost all the steps of natural language processing. (Steps of NLP – lexical analysis, syntactic analysis, semantic analysis, discourse analysis, and pragmatic analysis).

Consider the following sentence for an example;

"Raj tried to reach his friend on the mobile, but he didn't attend"

In this sentence, we have the presence of lexical, syntactic, and anaphoric ambiguities.

- **Lexical ambiguity** – The word "***tried***" means "***attempted***" not "***judged***" or "***tested***". Also, the word "***reach***" means "***establish communication***" not "***gain***" or "***pass***" or "***strive***".

- **Syntactic ambiguity** – The phrase "***on the mobile***" attached to "***reach***" and thus means "***using the mobile***". It is not attached to "***friend***".

- **Anaphoric ambiguity** – The anaphor "***he***" refers the "***friend***" not "***Raj***".

The following are the types of ambiguity with respect to natural language processing task;

Lexical ambiguity

Syntactic ambiguity

Semantic ambiguity

Anaphoric ambiguity

Pragmatic ambiguity

UNIT-2 Semantic Analysis

NLP - Word Sense Disambiguation.

LECTURE NO:-3

Word Sense Disambiguation.

Word Sense Disambiguation:

In Natural Language, the meaning of a word may vary as per its usage in sentences and the context of the text. Word Sense Disambiguation involves interpreting the meaning of a word based upon the context of its occurrence in a text.

For example, the word 'Bark' may mean 'the sound made by a dog' or 'the outermost layer of a tree.' Likewise, the word 'rock' may mean '*a stone*' or '*a genre of music*' – hence, the accurate meaning of the word is highly dependent upon its context and usage in the text.

Thus, the ability of a machine to overcome the ambiguity involved in identifying the meaning of a word based on its usage and context is called Word Sense Disambiguation.

Word Sense Disambiguation Word Sense Disambiguation (WSD) involves interpreting the meaning of a word based on the context of its occurrence in a text.

- Natural language is **ambiguous and polysemic**, and sometimes the same word can have different meanings depending on how it is used in the text.
- Word sense disambiguation is an **automated process of identifying** in which sense is a word used according to its context under elements of semantic analysis.

Need for Word Sense Disambiguation

- From a machine point of view, human text and human utterances from language and speech are open to **multiple interpretations** because words may have more than one meaning which is also called **lexical ambiguity**.
- It may also be because certain words such as quantifiers, modals, or negative operators may apply to different stretches of text called **scopal ambiguity**.
- It may also occur because the intended reference of pronouns or other referring expressions may be unclear, which is called **referential ambiguity**.

Ambiguity resolution is one of the frequently identified requirements for semantic analysis in NLP as the meaning of a word in natural language may vary as per its usage in sentences and the context of the text.

Classification of Methods for Word Sense Disambiguation

WSD approaches are categorized mainly into three types, Knowledge-based, Supervised, and Unsupervised methods.

- **Knowledge-based WSD**: Knowledge-based approaches are based on different knowledge sources such as machine-readable dictionaries or sense inventories, thesauri, etc.
    - Wordnet is the mostly used machine-readable dictionary in this research field.
    - Generally, four main types of knowledge-based methods are used under this category: LESK algorithm, Semantic similarity methods, selectional preferences, and Heuristic-based methods for WSD under elements of semantic analysis.
    - These algorithms generally give higher precision. These algorithms are overlap based, so they suffer from overlap sparsity and performance depends on dictionary definitions.
- **WSD using Supervised Approaches**: The supervised approaches applied to WSD systems use machine learning techniques from manually created sense annotated data.
    - Training set will be used for the classifier to learn and the training set which consists of examples related to the target word. The tags are manually created from the dictionary.
    - Supervised-based WSD algorithm generally gives better results than other approaches.
    - Decision rules, decision trees, Naive Bayes, Neural networks, instance-based learning methods, support vector machines, and ensemble-based methods are some algorithms used in this category.
    - This type of algorithm is better than the two approaches with respect to the implementation perspective. One other caveat with these algorithms is that they do not give satisfactory results for resource-scarce languages.**
- **WSD Using Unsupervised Approaches**: These methods do not depend on external knowledge sources or sense inventories, machine-readable dictionaries, or sense-annotated datasets.
    - These algorithms generally do not assign meaning to the words. Instead they discriminate the word meanings based on information found in unannotated corpora.
    - Unsupervised WSD has two types of distributional approaches. The first one is monolingual corpora, and the other one is translation equivalence based on parallel corpora.
    - The techniques are further categorized into two types, type-based and token-based approaches. The type-based approach disambiguates by clustering instances of a target word, and the token-based approach disambiguates by clustering the context of a target word using elements of semantic analysis.**

- Context clustering, Word clustering, Co-occurrence graph, and Spanning tree-based approaches are the main approaches under unsupervised WSD.
- There is no need for any sense inventory and sense annotated corpora in these approaches. These algorithms are **difficult to implement** and **performance is generally inferior** to that of the other two approaches.

EVALUATION OF WSD

The evaluation of WSD requires the following two inputs −

A Dictionary

The very first input for evaluation of WSD is dictionary, which is used to specify the senses to be disambiguated.

Test Corpus

Another input required by WSD is the high-annotated test corpus that has the target or correct-senses. The test corpora can be of two types &minsu;

- **Lexical sample** − This kind of corpora is used in the system, where it is required to disambiguate a small sample of words.
- **All-words** − This kind of corpora is used in the system, where it is expected to disambiguate all the words in a piece of running text.

APPROACHES AND METHODS TO WORD SENSE DISAMBIGUATION (WSD)

Approaches and methods to WSD are classified according to the source of knowledge used in word disambiguation.

Let us now see the four conventional methods to WSD −

Dictionary-based or Knowledge-based Methods

As the name suggests, for disambiguation, these methods primarily rely on dictionaries, treasures and lexical knowledge base. They do not use corpora evidences for disambiguation. The Lesk method is the seminal dictionary-based method introduced by Michael Lesk in 1986. The Lesk definition, on which the Lesk algorithm is based is **“measure overlap between sense definitions for all words in context”**. However, in 2000, Kilgarriff and Rosensweig gave the simplified Lesk definition as **“measure overlap between sense definitions of word and current context”**, which further means identify the correct sense for one word at a time. Here the current context is the set of words in surrounding sentence or paragraph.

Supervised Methods

For disambiguation, machine learning methods make use of sense-annotated corpora to train. These methods assume that the context can provide enough evidence on its own to disambiguate the sense. In

these methods, the words knowledge and reasoning are deemed unnecessary. The context is represented as a set of "features" of the words. It includes the information about the surrounding words also. Support vector machine and memory-based learning are the most successful supervised learning approaches to WSD. These methods rely on substantial amount of manually sense-tagged corpora, which is very expensive to create.

Semi-supervised Methods

Due to the lack of training corpus, most of the word sense disambiguation algorithms use semi-supervised learning methods. It is because semi-supervised methods use both labelled as well as unlabeled data. These methods require very small amount of annotated text and large amount of plain unannotated text. The technique that is used by semisupervised methods is bootstrapping from seed data.

Unsupervised Methods

These methods assume that similar senses occur in similar context. That is why the senses can be induced from text by clustering word occurrences by using some measure of similarity of the context. This task is called word sense induction or discrimination. Unsupervised methods have great potential to overcome the knowledge acquisition bottleneck due to non-dependency on manual efforts.

APPLICATIONS OF WORD SENSE DISAMBIGUATION (WSD)

Word sense disambiguation (WSD) is applied in almost every application of language technology.

Let us now see the scope of WSD −

Machine Translation

Machine translation or MT is the most obvious application of WSD. In MT, Lexical choice for the words that have distinct translations for different senses, is done by WSD. The senses in MT are represented as words in the target language. Most of the machine translation systems do not use explicit WSD module.

Information Retrieval (IR)

Information retrieval (IR) may be defined as a software program that deals with the organization, storage, retrieval and evaluation of information from document repositories particularly textual information. The system basically assists users in finding the information they required but it does not explicitly return the answers of the questions. WSD is used to resolve the ambiguities of the queries provided to IR system. As like MT, current IR systems do not explicitly use WSD module and they rely on the concept that user would type enough context in the query to only retrieve relevant documents.

Text Mining and Information Extraction (IE)

In most of the applications, WSD is necessary to do accurate analysis of text. For example, WSD helps intelligent gathering system to do flagging of the correct words. For example, medical intelligent system might need flagging of "illegal drugs" rather than "medical drugs"

Lexicography

WSD and lexicography can work together in loop because modern lexicography is corpusbased. With lexicography, WSD provides rough empirical sense groupings as well as statistically significant contextual indicators of sense.

DIFFICULTIES IN WORD SENSE DISAMBIGUATION (WSD)

Followings are some difficulties faced by word sense disambiguation (WSD) −

Differences between dictionaries

The major problem of WSD is to decide the sense of the word because different senses can be very closely related. Even different dictionaries and thesauruses can provide different divisions of words into senses.

Different algorithms for different applications

Another problem of WSD is that completely different algorithm might be needed for different applications. For example, in machine translation, it takes the form of target word selection; and in information retrieval, a sense inventory is not required.

Inter-judge variance

Another problem of WSD is that WSD systems are generally tested by having their results on a task compared against the task of human beings. This is called the problem of interjudge variance.

Word-sense discreteness

Another difficulty in WSD is that words cannot be easily divided into discrete submeanings.

UNIT-2 Semantic Analysis

NLP - Discourse Processing:

LECTURE NO:-4

Discourse Processing:

Discourse refers to any linguistic construction with multiple sentences. A disclosure is used in understanding and generating natural language. A variety of text mining applications can be supported by discourse processing, which is a collection of Natural Language Processing (NLP) tasks used to extract linguistic structures from texts at different levels. Identifying the conversational discourse's topic structure, coherence structure, coreference structure, and conversation structure is required for this. Together, these structures can guide information extraction, sentiment analysis, machine translation, question answering, essay scoring, text summarization, and thread recovery.

Discourse segmentation can be defined as the process of determining the types of structures that will be used in large amounts of discourse. Implementing discourse segmentation is difficult, but it is critical for applications such as information retrieval, text summarization, and information extraction. Discourse Integration is dependent on the sentences that come before it and also invokes the meaning of the sentences that come after it. Unsupervised Discourse Segmentation and Supervised Discourse Segmentation are popularly used algorithms for discourse segmentation.

Discourse in NLP is nothing but coherent groups of sentences. When we are dealing with Natural Language Processing, the provided language consists of structured, collective, and consistent groups of sentences, which are termed discourse in NLP. The relationship between words makes the training of the NLP model quite easy and more predictable than the actual results.

Discourse Analysis is extracting the meaning out of the corpus or text. Discourse Analysis is very important in Natural language Processing and helps train the NLP model better.

PRE-REQUISITES

Before learning about the Discourse in NLP, let us first learn some basics about the NLP itself.

- NLP stands for **Natural Language Processing**. In NLP, we perform the analysis and synthesis of the input, and the trained NLP model then predicts the necessary output.
- NLP is the backbone of technologies like Artificial Intelligence and Deep Learning. In basic terms, we can say that NLP is nothing but the computer program's ability to process and understand the provided human language.

INTRODUCTION

One of the primary challenges that we face in the world of Artificial Intelligence is processing Natural Language data by computers. We can even say that Natural Language Processing is quite a difficult issue in the field of AI. Now if we are talking about the major problem in Natural Language Processing, then we are talking about the processing of Discourse in NLP.

So, we can see that the real problem is the processing of the Discourse in NLP, and hence we need to work on it so that our model can be trained well, which will help in better processing of Natural Language data by the computers and hence the Artificial Intelligence can predict the desired result.

Now a question that comes to our mind is what is Discourse in NLP? Well, in simple terms, we can say that discourse in NLP is nothing but coherent groups of sentences. When we are dealing with Natural Language Processing, the provided language consists of structured, collective, and consistent groups of sentences, which are termed discourse in NLP. The relationship between words makes the training of the NLP model quite easy and more predictable than the actual results.

Discourse Analysis is extracting the meaning out of the corpus or text. Discourse Analysis is very important in Natural language Processing and helps train the NLP model better.

Let us now learn about the concept of coherence in the next section.

CONCEPT OF COHERENCE

Coherence in terms of Discourse in NLP means making sense of the utterances or making meaningful connections and correlations. There is a lot of connection between the coherence and the discourse structure (discussed in the next section). We use the property of good text, coherence, etc., to evaluate the quality of the output generated by the natural language processing generation system.

What are coherent discourse texts? Well, if we read a paragraph from a newspaper, we can see that the entire paragraph is interrelated; hence we can say that the discourse is coherence, but if we only combine

the newspaper headlines consecutively, then it is not a discourse, it is just a group of sentences that are also non-coherence.

Let us now learn about the two major properties of coherence, i.e., Coherence relation between utterances and Coherence relation between entities.

Coherence Relation Between Utterances

When we say that the discourses are coherent, then it simply means that the discourse has some sort of meaningful connection. The **coherent relation** tells us that there is some sort of connection present between the utterances.

Relationship Between Entities

If there is some kind of relationship between the entities, then we can also say that the discourse in NLP is coherent. So, the coherence between the entities is known as **entity-based coherence**.

DISCOURSE STRUCTURE

So far, we have discussed discourse and coherence, but we have not discussed the structure of the discourse in NLP. Let us now look at the structure that discourse in NLP must have. Now, the structure of the discourse depends on the type of segmentation applied to the discourse.

What is discourse segmentation ? Well, when we determine the types of structures for a large discourse, we term its segmentation. The segmentation is a difficult thing to implement, but it is very necessary as discourse segmentation is used in fields like :

- Information Retrieval,
- Text summarization,
- Information Extraction, etc.

ALGORITHMS FOR DISCOURSE SEGMENTATION

We have different algorithms for **Unsupervised Discourse Segmentation** and **Supervised Discourse Segmentation**. Let us now learn about the various algorithms used for discourse segmentation in this section.

Unsupervised Discourse Segmentation

The class of unsupervised segmentation is also termed or represented as linear segmentation. Let us take an example to understand this discourse segmentation better.

Suppose we have a text with us, and the task is to segment the text into various units of multi-paragraphs. In the multi-paragraphs, a single unit is going to represent a passage of the text.

Now the algorithm will take the help of cohesion (that we have discussed above), and the algorithm will classify the dependent texts and tie them together using some linguistic devices. In simpler terms,

unsupervised discourse segmentation means the classification and grouping up of similar texts with the help of coherent discourse in NLP.

The **unsupervised discourse segmentation** can also be performed with the help of lexicon cohesion. The lexicon cohesion indicates the relationship among similar units, for example, synonyms.

Supervised Discourse Segmentation

In the previous segmentation, there was no certain labeled segment boundary to separate the discourse segments. But in the supervised discourse segmentation, we only deal with the training data set having a labeled boundary. To differentiate or structure the discourse segments, we make use of cue words or discourse makers. These cue words or discourse maker works to signal the discourse structure. As there can be varied domains of discourse in NLP so, the cue words or discourse makers are domain specific.

TEXT COHERENCE

As we have previously discussed, the coherent discourse in NLP aims to find the coherence relation among the discourse text. Now, to find the structure in discourse, we use lexical repetition, but by using this lexical repetition, we cannot satisfy the conditions of coherent discourse. So, to prove such a kind of discourse relation, **Hebb** has proposed some solutions.

Suppose we have two kinds of related sentences, namely: S0 and S1.

Result

We can say that the second statement, i.e., S1 can be the cause of the first statement, i.e., S0. For example, Rahul is late. He will be punished.

In the above example, we can say that the first statement, S0, i.e., Rahul is late, has caused the second statement, i.e., S1, i.e., He will be punished.

Explanation

Similar to the result, We can say that the first statement, i.e., S0 can be the cause of the second statement, i.e., S1. For example, Rahul fought with his friend. He was drunk.

Parallel

By the term parallel, we mean that the assertion from the statement S0, i.e., p(a1, a2, …), and the assertion from the statement S1, i.e. p(b1, b2, …), the ai and bi is similar for all the values of I.

In simpler terms, it shows us that the sentences are parallel. For example, He wants food. She wants money. Both of the statements are parallel as there is a sense of want in both sentences.

Elaboration

Elaboration means that proposition P is inferring from both the assertions S0 and S1. For example, Rahul is from Delhi. Rohan is from Mumbai.

Occasion

The **occasion** takes place when the change in the state is inferred from the first assertion S0, the final state is inferred from the statement S1, and vice-versa. Let us take an example to understand the relationship occasion better. For example, Rahul took the money. he gave it to Rohan.

BUILDING HIERARCHICAL DISCOURSE STRUCTURE

In the previous section, we discussed how text coherence takes place. Let us now try to build a hierarchal discourse structure with the help of a group of statements. We generally create the hierarchical structure among the coherence relations to get the entire discourse in NLP.

Let us consider the following phrases and serially number them.

- **S1:**
  Rahul went to the bank to deposit money.
- **S2:**
  He then went to Rohan's shop.
- **S3 :**
  He wanted a phone.
- **S4 :**
  He did not have a phone.
- **S5:**
  He also wanted to buy a laptop from Rohan's shop.

Now the entire discourse can be represented using the below hierarchal discourse structure.

REFERENCE RESOLUTION

The extraction of the meaning or interpretation of the sentences of discourse is one of the most important tasks in natural language processing, and to do so, we first need to know what or who is the

entity that we are talking about. Reference resolution means understanding the type of entity that is being talked about.

By the term **reference**, we mean the linguistic expression that is used to denote an individual or an entity. For example, look at the below sentences.

- Rahul went to the farm.
- He cooked food.
- His farm was very big.

In the above sentences, Rahul, He, and His references. So, we can simply define the reference resolution as the task of determination of the entities that are being referred to by the linguistic expressions.

Let us now look at the various terminologies used in the reference resolution.

Terminology Used in Reference Resolution

- **Referring expression:**
  The NLP expression that performs the reference is termed a referring expression. For example, the passage that we have talked about in the above section is an example of the referring expression.
- **Referent:**
  Referent is the entity we have referred to. For example, in the above passage, Rahul is the referent.
- **Co-refer:**
  As the name suggests, Co-refer is a term used for an entity if two or more expressions are referring to the same entity. For example, Rahul and He is used for the same entity, i.e., Rahul.
- **Antecedent:**
  The term that has been licensed to use another term is termed antecedent. For example, in the above passage, Rahul is the antecedent of the reference He.
- **Anaphora & Anaphoric:**
  The referring expression is termed anaphoric. **Anaphora** & **Anaphoric** can be said to be the term or reference used for an entity that has previously been introduced in the same sentence.
- **Discourse model:**
  It is the model that has the overall representation of the entities that have been referred to in the discourse text. It also contains the relationship of the involved discourse in the NLP.

TYPES OF REFERRING EXPRESSIONS

As we have previously discussed, the NLP expression that performs the reference is termed a referring expression. We have mainly five types of referring expressions in Natural Language Processing. Let us discuss them one by one.

1. Indefinite Noun Phrases

Indefinite noun reference is a kind of reference that represents the entity that is new to the discourse context's hearer. To understand the indefinite noun phrase, let us take an example.

For example:
In the sentence : Rahul is doing some work., some is an indefinite noun phrase.

2. Definite Noun Phrases

A **definite noun reference** is a kind of reference that represents the entity that is not new to the discourse context's hearer. The discourse context's hearer can easily identify the definite noun reference. To understand the definite noun phrase, let us take an example.

For example:
In the sentence: Rahul loves reading the Times of India., the Times of India is an indefinite noun phrase.

3. Pronouns

Pronouns is a form of definite reference (its working is the same as we have learned in English grammar).

For example:
In the sentence, Rahul learned as much as he could. Here, he is the pronoun that is referring to the noun Rahul.

4. Demonstratives

The **demonstratives** are also used to demonstrate the nouns but they behave differently than the simple pronouns.

For example, that, this, these, and those, are some examples of demonstratives.

5. Names

Names can be the name of the person, location, organization, etc. So, it is the simplest form of referring to the expressions.

For example, in the above examples, Rahul is the name referring expression.

REFERENCE RESOLUTION TASKS

To resolve the reference, we can use the two resolution tasks. Let us discuss them one by one.

1. Co-reference Resolution

In the **Co-reference Resolution**, the main aim is to find the referring expression from the provided text that refers to the same entity. In a discourse in NLP, Co-refer is a term used for an entity if two or more expressions are referring to the same entity.

For example, Rahul and He is used for the same entity i.e., Rahul.

The **Co-reference Resolution** can be simply termed as finding the relevant co-refer expressions among the provided discourse text. Let us take an example for more clarity.

For example, Rahul went to the farm. He cooked food. In this example, Rahul and He is the referring expressions.

We have some sort of constraints present on the Co-reference Resolution. Let us learn about the constraint.

Constraint on Co-reference Resolution:

In the English language, we have many pronouns. If we are using the pronouns he and she, then we can easily resolve it. But if we are using the pronoun it, the resolution can be tricky, and if we have a set of co-referring expressions, then it becomes more complex to resolve it. In simpler terms, if we are using the it pronoun, then the exact determination of the referred noun is complex.

2. Pronominal Anaphora Resolution

By the terms **Pronominal Anaphora Resolution**, we are aiming to find the antecedent for the current single pronoun.

For example, in the passage - Rahul went to the farm. He cooked food., Rahul is the antecedent of the reference He.

CONCLUSION

- **Discourse in NLP** is nothing but coherent groups of sentences. When we are dealing with Natural Language Processing, the provided language consists of structured, collective, and consistent groups of sentences, which are termed discourse in NLP.
- **Discourse Analysis** is very important in Natural language Processing and helps train the NLP model better.
- **Coherence** in terms of Discourse in NLP means making sense between the utterances or making meaningful connections and correlations. We use the property of good text, coherence, etc. to evaluate the quality of the output generated by the natural language processing generation system.
- The extraction of the meaning or interpretation of the sentences of discourse is one of the most important tasks in natural language processing, and to do so, we first need to know what or who is the entity that we are talking about.
- **Indefinite noun reference** is a kind of reference that represents the entity that is new to the discourse context's hearer.
- **Definite noun reference** is a kind of reference that represents the entity that is not new to the discourse context's hearer. The discourse context's hearer can easily identify the definite noun reference.
- In the **Co-reference Resolution**, the main aim is to find the referring expression from the provided text that refers to the same entity. By the terms **Pronominal Anaphora Resolution**, we are aiming to find the antecedent for the current single pronoun.

UNIT-2 Semantic Analysis

NLP - cohesion, Reference Resolution,

LECTURE NO:-6

cohesion, Reference Resolution,

The most difficult problem of AI is to process the natural language by computers or in other words *natural language processing* is the most difficult problem of artificial intelligence. If we talk about the major problems in NLP, then one of the major problems in NLP is discourse processing − building theories and models of how utterances stick together to form **coherent discourse**. Actually, the language always consists of collocated, structured and coherent groups of sentences rather than isolated and unrelated sentences like movies. These coherent groups of sentences are referred to as discourse.

CONCEPT OF COHERENCE

Coherence and discourse structure are interconnected in many ways. Coherence, along with property of good text, is used to evaluate the output quality of natural language generation system. The question that arises here is what does it mean for a text to be coherent? Suppose we collected one sentence from every page of the newspaper, then will it be a discourse? Of-course, not. It is because these sentences do not exhibit coherence. The coherent discourse must possess the following properties −

Coherence relation between utterances

The discourse would be coherent if it has meaningful connections between its utterances. This property is called coherence relation. For example, some sort of explanation must be there to justify the connection between utterances.

Relationship between entities

Another property that makes a discourse coherent is that there must be a certain kind of relationship with the entities. Such kind of coherence is called entity-based coherence.

DISCOURSE STRUCTURE

An important question regarding discourse is what kind of structure the discourse must have. The answer to this question depends upon the segmentation we applied on discourse. Discourse segmentations may be defined as determining the types of structures for large discourse. It is quite difficult to implement discourse segmentation, but it is very important for **information retrieval, text summarization and information extraction** kind of applications.

ALGORITHMS FOR DISCOURSE SEGMENTATION

In this section, we will learn about the algorithms for discourse segmentation. The algorithms are described below −

Unsupervised Discourse Segmentation

The class of unsupervised discourse segmentation is often represented as linear segmentation. We can understand the task of linear segmentation with the help of an example. In the example, there is a task of segmenting the text into multi-paragraph units; the units represent the passage of the original text. These algorithms are dependent on cohesion that may be defined as the use of certain linguistic devices to tie the textual units together. On the other hand, lexicon cohesion is the cohesion that is indicated by the relationship between two or more words in two units like the use of synonyms.

Supervised Discourse Segmentation

The earlier method does not have any hand-labeled segment boundaries. On the other hand, supervised discourse segmentation needs to have boundary-labeled training data. It is very easy to acquire the same. In supervised discourse segmentation, discourse marker or cue words play an important role. Discourse marker or cue word is a word or phrase that functions to signal discourse structure. These discourse markers are domain-specific.

TEXT COHERENCE

Lexical repetition is a way to find the structure in a discourse, but it does not satisfy the requirement of being coherent discourse. To achieve the coherent discourse, we must focus on coherence relations in specific. As we know that coherence relation defines the possible connection between utterances in a discourse. Hebb has proposed such kind of relations as follows −

We are taking two terms $\mathbf{S_0}$ and $\mathbf{S_1}$ to represent the meaning of the two related sentences −

Result

It infers that the state asserted by term $\mathbf{S_0}$ could cause the state asserted by $\mathbf{S_1}$. For example, two statements show the relationship result: Ram was caught in the fire. His skin burned.

Explanation

It infers that the state asserted by $\mathbf{S_1}$ could cause the state asserted by $\mathbf{S_0}$. For example, two statements show the relationship − Ram fought with Shyam's friend. He was drunk.

Parallel

It infers p(a1,a2,…) from assertion of $\mathbf{S_0}$ and p(b1,b2,…) from assertion $\mathbf{S_1}$. Here ai and bi are similar for all i. For example, two statements are parallel − Ram wanted car. Shyam wanted money.

Elaboration

It infers the same proposition P from both the assertions − **S_0** and **S_1** For example, two statements show the relation elaboration: Ram was from Chandigarh. Shyam was from Kerala.

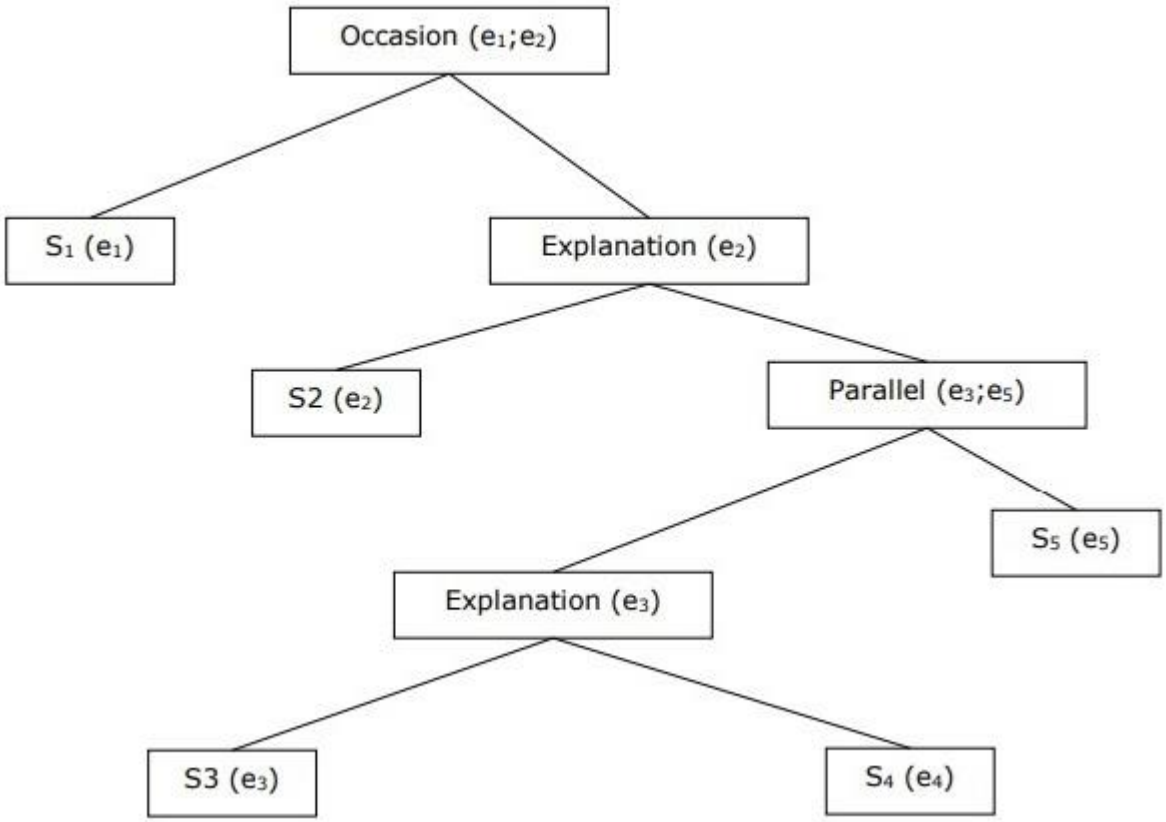
Occasion

It happens when a change of state can be inferred from the assertion of **S_0**, final state of which can be inferred from **S_1** and vice-versa. For example, the two statements show the relation occasion: Ram picked up the book. He gave it to Shyam.

BUILDING HIERARCHICAL DISCOURSE STRUCTURE

The coherence of entire discourse can also be considered by hierarchical structure between coherence relations. For example, the following passage can be represented as hierarchical structure −

- **S_1** − Ram went to the bank to deposit money.
- **S_2** − He then took a train to Shyam's cloth shop.
- **S_3** − He wanted to buy some clothes.
- **S_4** − He do not have new clothes for party.
- **S_5** − He also wanted to talk to Shyam regarding his health



REFERENCE RESOLUTION

Interpretation of the sentences from any discourse is another important task and to achieve this we need to know who or what entity is being talked about. Here, interpretation reference is the key element. **Reference** may be defined as the linguistic expression to denote an entity or individual. For example, in the passage, Ram, the manager of ABC bank, saw his friend Shyam at a shop. He went to meet him, the linguistic expressions like Ram, His, He are reference.

On the same note, **reference resolution** may be defined as the task of determining what entities are referred to by which linguistic expression.

TERMINOLOGY USED IN REFERENCE RESOLUTION

We use the following terminologies in reference resolution −

- **Referring expression** − The natural language expression that is used to perform reference is called a referring expression. For example, the passage used above is a referring expression.
- **Referent** − It is the entity that is referred. For example, in the last given example Ram is a referent.
- **Corefer** − When two expressions are used to refer to the same entity, they are called corefers. For example, ***Ram*** and ***he*** are corefers.
- **Antecedent** − The term has the license to use another term. For example, ***Ram*** is the antecedent of the reference ***he***.
- **Anaphora & Anaphoric** − It may be defined as the reference to an entity that has been previously introduced into the sentence. And, the referring expression is called anaphoric.
- **Discourse model** − The model that contains the representations of the entities that have been referred to in the discourse and the relationship they are engaged in.

TYPES OF REFERRING EXPRESSIONS

Let us now see the different types of referring expressions. The five types of referring expressions are described below −

Indefinite Noun Phrases

Such kind of reference represents the entities that are new to the hearer into the discourse context. For example − in the sentence Ram had gone around one day to bring him some food − some is an indefinite reference.

Definite Noun Phrases

Opposite to above, such kind of reference represents the entities that are not new or identifiable to the hearer into the discourse context. For example, in the sentence - I used to read The Times of India – The Times of India is a definite reference.

Pronouns

It is a form of definite reference. For example, Ram laughed as loud as he could. The word **he** represents pronoun referring expression.

Demonstratives

These demonstrate and behave differently than simple definite pronouns. For example, this and that are demonstrative pronouns.

Names

It is the simplest type of referring expression. It can be the name of a person, organization and location also. For example, in the above examples, Ram is the name-refereeing expression.

REFERENCE RESOLUTION TASKS

The two reference resolution tasks are described below.

Coreference Resolution

It is the task of finding referring expressions in a text that refer to the same entity. In simple words, it is the task of finding corefer expressions. A set of coreferring expressions are called coreference chain. For example - He, Chief Manager and His - these are referring expressions in the first passage given as example.

Constraint on Coreference Resolution

In English, the main problem for coreference resolution is the pronoun it. The reason behind this is that the pronoun it has many uses. For example, it can refer much like he and she. The pronoun it also refers to the things that do not refer to specific things. For example, It's raining. It is really good.

Pronominal Anaphora Resolution

Unlike the coreference resolution, pronominal anaphora resolution may be defined as the task of finding the antecedent for a single pronoun. For example, the pronoun is his and the task of pronominal anaphora resolution is to find the word Ram because Ram is the antecedent.

Knowledge Representation, reasoning.

WHAT IS KNOWLEDGE REPRESENTATION?

Humans are best at understanding, reasoning, and interpreting knowledge. Human knows things, which is knowledge and as per their knowledge they perform various actions in the real world. **But how machines do all these things comes under knowledge representation and reasoning**. Hence we can describe Knowledge representation as following:

- Knowledge representation and reasoning (KR, KRR) is the part of Artificial intelligence which concerned with AI agents thinking and how thinking contributes to intelligent behavior of agents.
- It is responsible for representing information about the real world so that a computer can understand and can utilize this knowledge to solve the complex real world problems such as diagnosis a medical condition or communicating with humans in natural language.
- It is also a way which describes how we can represent knowledge in artificial intelligence. Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

WHAT TO REPRESENT:

Following are the kind of knowledge which needs to be represented in AI systems:

- **Object:** All the facts about objects in our world domain. E.g., Guitars contains strings, trumpets are brass instruments.
- **Events:** Events are the actions which occur in our world.
- **Performance:** It describe behavior which involves knowledge about how to do things.
- **Meta-knowledge:** It is knowledge about what we know.
- **Facts:** Facts are the truths about the real world and what we represent.
- **Knowledge-Base:** The central component of the knowledge-based agents is the knowledge base. It is represented as KB. The Knowledgebase is a group of the Sentences (Here, sentences are used as a technical term and not identical with the English language).

Knowledge: Knowledge is awareness or familiarity gained by experiences of facts, data, and situations. Following are the types of knowledge in artificial intelligence:

TYPES OF KNOWLEDGE

Following are the various types of knowledge:



1. Declarative Knowledge:

- Declarative knowledge is to know about something.
- It includes concepts, facts, and objects.
- It is also called descriptive knowledge and expressed in declarativesentences.
- It is simpler than procedural language.

2. Procedural Knowledge

- It is also known as imperative knowledge.
- Procedural knowledge is a type of knowledge which is responsible for knowing how to do something.
- It can be directly applied to any task.
- It includes rules, strategies, procedures, agendas, etc.
- Procedural knowledge depends on the task on which it can be applied.

3. Meta-knowledge:

- Knowledge about the other types of knowledge is called Meta-knowledge.

4. Heuristic knowledge:

- Heuristic knowledge is representing knowledge of some experts in a filed or subject.
- Heuristic knowledge is rules of thumb based on previous experiences, awareness of approaches, and which are good to work but not guaranteed.

5. Structural knowledge:

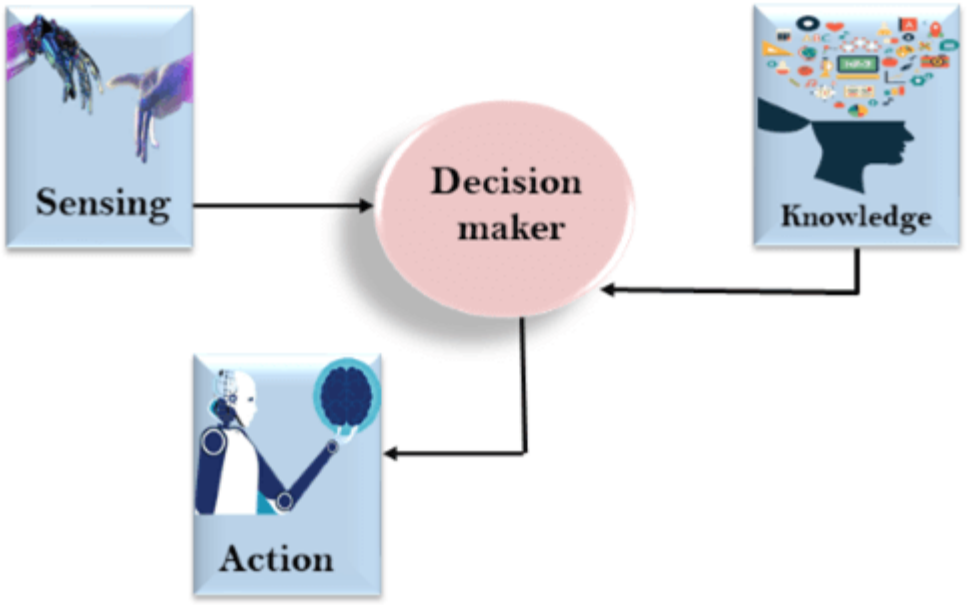
- Structural knowledge is basic knowledge to problem-solving.
- It describes relationships between various concepts such as kind of, part of, and grouping of something.
- It describes the relationship that exists between concepts or objects.

THE RELATION BETWEEN KNOWLEDGE AND INTELLIGENCE:

Knowledge of real-worlds plays a vital role in intelligence and same for creating artificial intelligence. Knowledge plays an important role in demonstrating intelligent behavior in AI agents. An agent is only able to accurately act on some input when he has some knowledge or experience about that input.

Let's suppose if you met some person who is speaking in a language which you don't know, then how you will able to act on that. The same thing applies to the intelligent behavior of the agents.

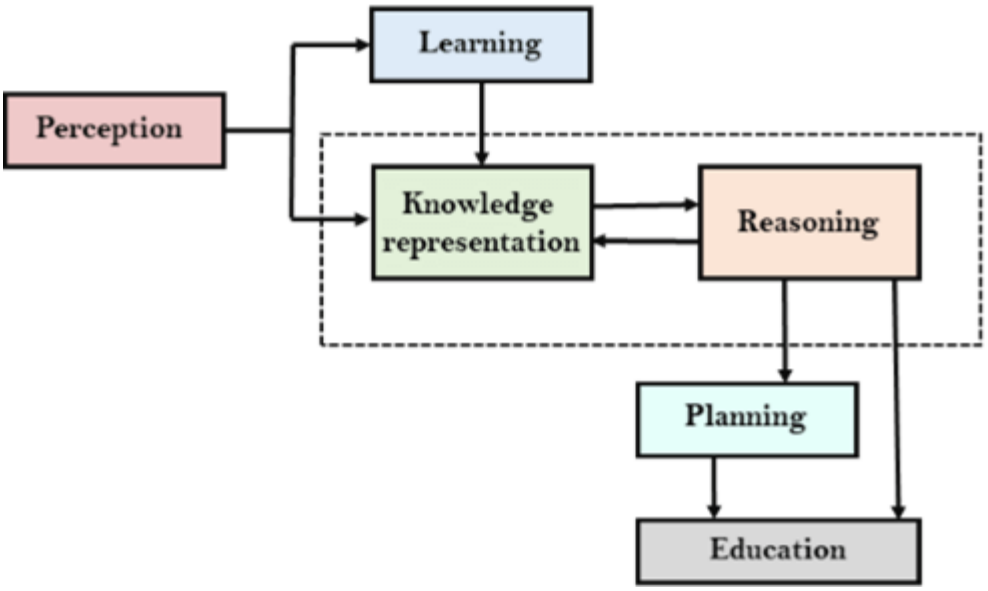
As we can see in below diagram, there is one decision maker which act by sensing the environment and using knowledge. But if the knowledge part will not present then, it cannot display intelligent behavior.



AI KNOWLEDGE CYCLE:

An Artificial intelligence system has the following components for displaying intelligent behavior:

- Perception
- Learning
- Knowledge Representation and Reasoning
- Planning
- Execution



The above diagram is showing how an AI system can interact with the real world and what components help it to show intelligence. AI system has Perception component by which it retrieves information from its environment. It can be visual, audio or another form of sensory input. The learning component is responsible for learning from data captured by Perception comportment. In the complete cycle, the main components are knowledge representation and Reasoning. These two components are involved in showing the intelligence in machine-like humans. These two components are independent with each other but also coupled together. The planning and execution depend on analysis of Knowledge representation and reasoning.

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APPROACHES TO KNOWLEDGE REPRESENTATION:

There are mainly four approaches to knowledge representation, which are givenbelow:

1. SIMPLE RELATIONAL KNOWLEDGE:

- It is the simplest way of storing facts which uses the relational method, and each fact about a set of the object is set out systematically in columns.
- This approach of knowledge representation is famous in database systems where the relationship between different entities is represented.
- This approach has little opportunity for inference.

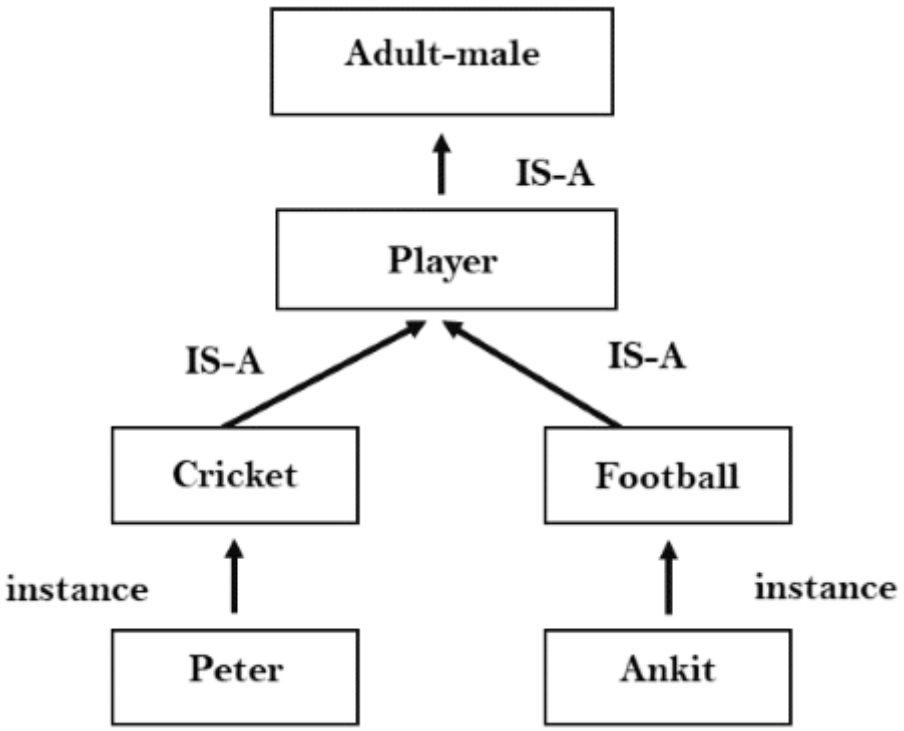
Example: The following is the simple relational knowledge representation.

Player	Weight	Age
Player1	65	23
Player2	58	18
Player3	75	24

2. INHERITABLE KNOWLEDGE:

- In the inheritable knowledge approach, all data must be stored into a hierarchy of classes.
- All classes should be arranged in a generalized form or a hierarchal manner.
- In this approach, we apply inheritance property.
- Elements inherit values from other members of a class.

- This approach contains inheritable knowledge which shows a relation between instance and class, and it is called instance relation.
- Every individual frame can represent the collection of attributes and its value.
- In this approach, objects and values are represented in Boxed nodes.
- We use Arrows which point from objects to their values.
- **Example:**



3. INFERENTIAL KNOWLEDGE:

- Inferential knowledge approach represents knowledge in the form of formal logics.
- This approach can be used to derive more facts.
- It guaranteed correctness.
- **Example:** Let's suppose there are two statements:
  a. Marcus is a man
  b. All men are mortal
  Then it can represent as;

  man(Marcus)
  ∀x = man (x) ----------> mortal (x)s

4. PROCEDURAL KNOWLEDGE:

- Procedural knowledge approach uses small programs and codes which describes how to do specific things, and how to proceed.
- In this approach, one important rule is used which is **If-Then rule**.
- In this knowledge, we can use various coding languages such as **LISP language** and **Prolog language**.
- We can easily represent heuristic or domain-specific knowledge using this approach.
- But it is not necessary that we can represent all cases in this approach.

REQUIREMENTS FOR KNOWLEDGE REPRESENTATION SYSTEM:

A good knowledge representation system must possess the following properties.

1. **1.Representational Accuracy:** KR system should have the ability to represent all kind of required knowledge.
2. **2. Inferential Adequacy:** KR system should have ability to manipulate the representational structures to produce new knowledge corresponding to existing structure.
3. **3.Inferential Efficiency:** The ability to direct the inferential knowledge mechanism into the most productive directions by storing appropriate guides.
4. **4. Acquisitional efficiency-** The ability to acquire the new knowledge easily using automatic methods.